$(BC)A = (BC)(C^{-1}B^{-1}) = ((BC)C^{-1})B^{-1}$ (Theorem 2(a))

$= (B(CC^{-1}))B^{-1}$ (Theorem 2(a)) $= (BI)B^{-1}$ (By Defn. of inverse)

$= BB^{-1}$ (Theorem 2(e)) $= I$ (By defn. of inverse). □

Now we shall show that BC is the unique solution of both the equations.

$AX = I \Rightarrow (C^{-1}B^{-1})X = I \Rightarrow C((C^{-1}B^{-1})X) = CI$

$\Rightarrow C(C^{-1}(B^{-1}X)) = CI$ (Theorem 2(a))

$\Rightarrow (CC^{-1})(B^{-1}X) = CI$ (Theorem 2(a))

$\Rightarrow I(B^{-1}X) = CI$ (By defn. of inverse)

$\Rightarrow B^{-1}X = C$ (Theorem 2(e))

$\Rightarrow B(B^{-1}X) = BC \Rightarrow (BB^{-1})X = BC$ (Theorem 2(a)) $\Rightarrow IX = BC$ (Defn. of inverse)

$\Rightarrow \boxed{X = BC}$ (Theorem 2(e))

~~$XA = I \Rightarrow X(C^{-1}B^{-1}) = I \Rightarrow C(C^{-1}B^{-1}) = CI \Rightarrow (CC^{-1})B^{-1} = CI$ (Theorem 2(a))~~

~~$\Rightarrow$~~ $(X(C^{-1}B^{-1}))B = IB$

$XA = I \Rightarrow X(C^{-1}B^{-1}) = I \Rightarrow (XC^{-1})(B^{-1}B) = IB$ (Theorem 2(a))

$\Rightarrow ((XC^{-1})B^{-1})B = IB$ (Theorem 2(a)) $\Rightarrow XC^{-1} = B$ (Theorem 2(e))

$\Rightarrow (XC^{-1})I = IB$ (By the defn. of inverse) $\Rightarrow X(C^{-1}C) = BC$ (Theorem 2(a)) $\Rightarrow XI = BC$ (by the defn.

$\Rightarrow (XC^{-1})C = BC$ (Theorem 2(e))

of inverse) $\Rightarrow \boxed{X = BC}$ (Theorem 2(e))

By the defn. of invertible matrix, A is invertible □

17) Solve the equation $AB = BC$ for A, assuming that A, B and C are square, and B is invertible.

Solution: By the defn. of invertible matrix, there exists a unique matrix

$BB^{-1} = B^{-1}B = I$

$AB = BC \Rightarrow (AB)B^{-1} = BCB^{-1} \Rightarrow A(BB^{-1}) = BCB^{-1}$ (Theorem 2(a))

$\Rightarrow AI = BCB^{-1}$ (By the defn. of inverse) $\Rightarrow \boxed{A = BCB^{-1}}$ (Theorem 2(e))